

Zere Bekzhanova

SUSTAINABILITY ENGINEER · EMBODIED CARBON · LCA · BUILDING PERFORMANCE

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Relocating to the Netherlands · open to North Italy (Milan) · EU work-authorised

Building engineer specialising in embodied carbon, life-cycle assessment, and building-performance simulation. MASc (University of Toronto, 2025) in material use and embodied-GHG assessment for building renovation. LEED Green Associate and One Click LCA Construction LCA Specialist, with hands-on command of IES VE, EnergyPlus, OpenLCA, One Click LCA, Revit, and Python. Peer-reviewed author (ETH Zurich SBE25; two Q1 journals). Seeking sustainability-consulting, energy-modelling, and decarbonisation roles.

EXPERIENCE

Graduate Research Assistant — Embodied Carbon & LCA Sep 2023 – Oct 2025

University of Toronto · Centre for the Sustainable Built Environment (Prof. Shoshanna Saxe)

- Built a 3D multi-phase Revit model with material take-offs to quantify cradle-to-gate GHG emissions across renovation scenarios.
- Developed Python analysis pipelines for GHG quantification and comparative scenario assessment; results now under peer review.
- Presented at the ETH Zurich SBE25 Conference (2025) and the CSBE Embodied GHG Symposium (2024).
- Won the Climate Impacts Hackathon 2024 — Python + QGIS pipeline siting 8 cooling centres in Toronto's most heat-vulnerable wards.

System / Business Analyst Feb 2026 – Present

ZAAM Construction · Almaty, Kazakhstan

- Designed KPI tracking and reporting frameworks for construction project operations.
- Structured internal processes and documentation from scratch to improve delivery consistency.

Undergraduate Research Assistant — Sustainable Materials Sep 2019 – Jan 2022

Nazarbayev University · School of Engineering & Digital Sciences (Prof. Shazim Ali Memon)

- Researched self-sensing concrete and bagasse-based concrete; co-authored two Q1 journal papers.

Engineering Intern 2019 – 2022

Schneider Electric · BI Group · KAZGOR Design Academy — Kazakhstan

- Electrical cabinet design (Schneider), construction cost estimation (BI Group), and load-resistance analysis to SNIP codes (KAZGOR).

EDUCATION

MASc, Civil Engineering 2023 – 2025

University of Toronto · Centre for the Sustainable Built Environment — GPA 3.68/4.00, fully funded

MS Building & Architectural Engineering (first year, transferred) 2022 – 2023

Politecnico di Milano · full DSU scholarship · avg 27.38/30 · English-medium

BEng, Civil Engineering 2017 – 2021

Nazarbayev University · full scholarship

TECHNICAL SKILLS

Energy modelling: IES VE, EnergyPlus, OpenStudio, Climate Studio

LCA / embodied carbon: One Click LCA, OpenLCA

BIM / CAD / 3D: Revit, AutoCAD, Rhino

Data: Python (pandas, NumPy), Excel, QGIS

Frameworks: LEED, BREEAM, WELL

LEED GREEN ASSOCIATE

CONSTRUCTION LCA SPECIALIST

PUBLICATIONS & LANGUAGES

SBE25, ETH Zurich (2025) — embodied GHG in refurbishment, IOP Conf. Series

Buildings 12(10), 2022 — recycled concrete aggregate ITZ review

Nanomaterials 11(9), 2021 — self-sensing cementitious composites

Languages: Russian & Kazakh (native), English (IELTS 8.5), Italian (pre-intermediate)